

Unequal Recording-Layer Thickness with Readback Signal Rescaling and MLP Detection in Dual-Layer BPMR Systems

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This work proposes a dual-layer bit-patterned magnetic recording (DL-BPMR) system with unequal recording-layer thicknesses to improve bottom-layer detection under severe inter-layer interference (ILI). The thicker top layer enables reliable top-layer detection with a threshold detector, while a rescaling process based on top-layer decisions reduces ILI ambiguity in bottom-layer detection. The rescaled signal is then processed by both a partial-response maximum-likelihood (PRML) and a multilayer perceptron (MLP) detector. At a user density of 4.0 Tb/in², the proposed method achieves the best performance, achieving a BER below 10⁻⁵ at 20 dB and outperforming one-dimensional PRML rescaling by approximately 2 dB at a BER of 10⁻⁴. These results demonstrate the effectiveness of combining unequal layer thickness, readback signal rescaling, and MLP detection for high-density DL-BPMR.

Index Terms—Dual-layer bit-patterned magnetic recording, inter-layer interference, signal rescale, multilayer perceptron.

I. INTRODUCTION

The relentless demand for higher areal density (AD) in digital storage has driven conventional magnetic recording toward its physical limit, known as the superparamagnetic limit. To address this challenge, bit-patterned magnetic recording (BPMR) has emerged as a promising solution by employing lithographically defined, isolated magnetic islands that maintain thermal stability at ultra-high recording densities [1]. Furthermore, to extend the AD beyond the capability of single-layer magnetic recording, dual-layer BPMR (DL-BPMR) architectures have been proposed in which stacking two independent magnetic layers within a single medium effectively doubles the recording density [2].

However, the primary bottleneck in DL-BPMR systems lies in retrieving data from the bottom layer, as the reader is farther away. As a result, reliable detection of the bottom-layer data from the readback signal, which is a superposition of the magnetic responses from both layers, becomes more challenging. In addition, at high ADs, the recording environment becomes increasingly prone to interference. Under these conditions, the conventional two-dimensional (2D) impairments, namely inter-symbol interference (ISI) and inter-track interference (ITI), are further aggravated by inter-layer interference (ILI) arising from the two layers [3], thereby degrading the overall magnetic recording performance.

To address this issue, this study proposes a DL-BPMR architecture in which the top recording layer is thicker than the bottom layer, as illustrated in Fig. 1(a). This structural difference produces a distinct amplitude difference that improves the separation of the top-layer signal, as shown in Fig. 1(b), thereby allowing signal decoupling before processing of the top recording layer. As a result, the top-layer data can be detected more reliably using a simple threshold detector. Conversely, for bottom-layer detection, a rescaling method [4] is used together with the top-layer estimator to reconstruct and extract an enhanced bottom-layer signal. This reconstructed signal is then fed into a neural network (NN), specifically a Multilayer Perceptron (MLP), to detect the bottom-layer recording sequence.

Finally, the system performance is evaluated in terms of bit-error rate (BER) versus signal-to-noise ratio (SNR) and compared with that of a single-layer magnetic recording

(SLMR) system at an AD of 4.0 Tb/in² (2.0 Tb/in² per layer), confirming its effectiveness for high-density magnetic recording.

II. PROPOSED SYSTEM

Consider the channel model shown in Fig. 2. The user bits a_k^t and a_k^b are separately recorded on the top layer (TL) and bottom layer (BL), respectively. In this work, the head-media spacing (HMS) for the TL and BL was fixed at 3 nm and 7 nm, respectively. The thicknesses of the soft magnetic underlayer (SUL), interlayer 1 (IL1), BL, interlayer 2 (IL2), and TL were fixed at 300, 10, 2, 1, and 3 nm, respectively. The diameter (D) of the bit islands was 10 nm.

During the readback process, a single read head is positioned above the center of the top layer. The generated readback signal is affected by intersymbol interference (ISI), intertrack interference (ITI), and interlayer interference (ILI). This readback signal is modeled as the 2D convolution of the magnetization patterns of both bit-patterned media layers with the read head's sensitivity response. Thus, the received signal is expressed as the superposition of the signals from both layers: $r(t) = r^t(t) + r^b(t) + n(t)$ where $n(t)$ denotes additive white Gaussian noise (AWGN) with zero mean and variance σ^2 . The noisy readback signal is then sampled at the rate of T_x , where T_x is the bit period. The readback sequence is expressed as $r_k = r(t)|_{t=kT_x}$. Next, r_k is passed to a threshold detector to estimate the top-layer user bits, $\hat{a}_k^t$.

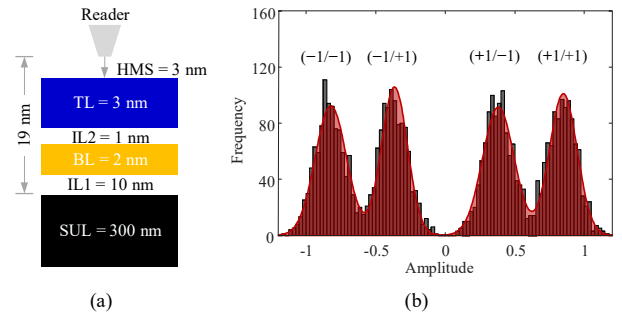


Fig. 1. Geometry of (a) proposed DL-BPMR architecture with unequal recording-layer thickness (b) readback signal amplitude distributions for each pair of recorded bits before rescaling at SNR = 12 dB.

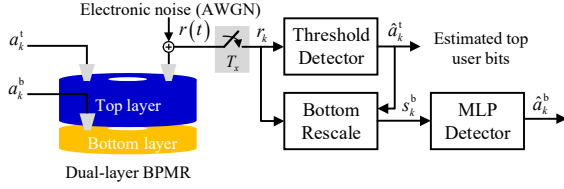


Fig 2. The proposed DL-BPMR channel model.

Once $\hat{a}_k^t$ is obtained from the threshold detector, the rescaling process uses these decisions to divide the received samples r_k into two groups. Samples corresponding to $\hat{a}_k^t \geq 0$ are assigned to the positive group, whereas samples with $\hat{a}_k^t < 0$ are assigned to the negative group.

Since the unequal layer thicknesses cause the two groups to occupy different amplitude ranges, the negative group is remapped onto the amplitude range of the positive group using min-max normalization:

$$S_k^b = [(r_k - r_{\min}^{\text{neg}}) / (r_{\max}^{\text{neg}} - r_{\min}^{\text{neg}})] \times (r_{\max}^{\text{pos}} - r_{\min}^{\text{pos}}) + r_{\min}^{\text{pos}}, \quad (1)$$

where S_k^b is the rescaled bottom-layer signal, $r_{\min}^{\text{neg}}$ and $r_{\max}^{\text{neg}}$ are the minimum and maximum amplitudes of the negative group, $r_{\min}^{\text{pos}}$ and $r_{\max}^{\text{pos}}$ are the minimum and maximum amplitudes of the positive group, respectively. This operation aligns both groups within the same amplitude range, thereby collapsing the original four-peak distribution, as shown in Fig. 1(b), into a two-peak distribution, as shown in Fig. 3, and significantly reducing ILI ambiguity before detection.

The rescaled signal S_k^b is then fed into an MLP detector to estimate the final bottom-layer bits, denoted by $\hat{a}_k^b$. The MLP architecture used in this work consists of an input layer, five hidden layers, and an output layer. The input layer contains 22 nodes, comprising 11 bits of estimated top-layer bits and 11 samples from the readback sequences. Each hidden layer contains 512 nodes, and the activation function for each node is the leaky rectified linear unit (Leaky ReLU), defined as,

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0.01x & \text{if } x < 0 \end{cases}. \quad (2)$$

In the output layer, the softmax activation function is employed to generate the final decision for the estimated bottom-layer user bits.

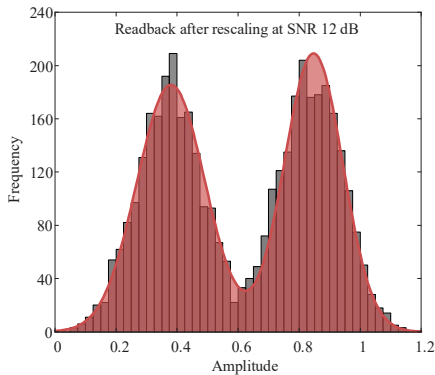


Fig 3. Readback signal amplitude distributions after rescaling at SNR = 12 dB.

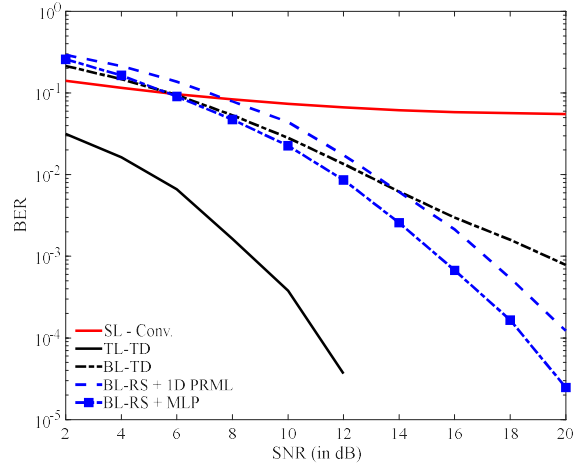


Fig 4. BER performance of the proposed system compared with various other recording systems. TL = top layer, BL = bottom layer.

III. SIMULATION RESULTS

We evaluate the performance of our proposed system in terms of BER. Fig. 4 presents the BER performance as a function of SNR for various detection schemes. Single-layer conventional (SL-Conv.) serves as the baseline. The top-layer-based threshold detection (TL-TD) achieved a BER of 10^{-4} at approximately 11 dB, confirming the reliable detection of top-layer data. In contrast, bottom threshold detection (BL-TD) performed poorly, indicating that direct threshold detection is insufficient for bottom-layer detection due to ILI. With the proposed rescaling process, bottom rescaling with 1D partial response maximum likelihood (PRML) (BL-RS + 1D PRML) provided an improvement over BL-TD when the SNR was greater than 14 dB. Among the evaluated methods, bottom rescaling with MLP (BL-RS + MLP) achieved the best performance, with a BER below 10^{-5} at an SNR of 20 dB. In addition, it outperformed BL-RS + 1D PRML by approximately 2 dB at a BER of 10^{-4} . These results demonstrate the effectiveness of combining asymmetric layer thickness, rescaling, and MLP-based detection.

ACKNOWLEDGMENTS

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